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The Impact of Physical Activity on Undergraduate Student Happiness

Abstract

This study aimed to understand the relationship between self-reported physical activity and happiness measures of undergraduate students at Washington University in St. Louis and assess whether the relationship changes depending on the level of physical activity (High, Moderate, or Low) and the setting in which it was undertaken (Leisure Time, Commuting, or Occupational physical activity). A descriptive and cross-sectional study was conducted. Self-reported physical activity under the three settings and happiness measures were collected using a survey questionnaire. The sample comprised thirty-five undergraduate students. An association between students' physical activity level and their happiness was observed across all settings – the higher the physical activity, the higher the average happiness score – and the difference in average happiness scores between High, Moderate, and Low activity levels was found to be statistically significant for Total physical activity, which was computed as the sum of physical activity levels across the three settings.

1. Introduction

Data from the Healthy Minds Network, a research organization that contributes to adolescent and young adult mental health research, suggests that the mental health of college students is deteriorating. Results from their annual surveys show that the significant metrics tracked by the survey – depression, major depression, suicidal thoughts, anxiety disorder, and eating disorder – have gone from already concerning levels in 2018-2019 to worse levels in 2021-2022 (Cornett 2023).

A large body of literature has documented the beneficial effects of physical activity in mitigating negative mental health conditions such as depression and anxiety. Yet, as noted by the constitution of the World Health Organization, “health is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity” (“Mental Health” 2014). So, understanding the effects of physical activity on the positive attributes of mental health, such as happiness, life satisfaction, and healthy self-esteem, is important. A comprehensive review study by University of Michigan researchers found only twenty-three studies since 1980 that focused exclusively on the relationship between physical activity and happiness (Zhang and Chen 2018). Many of those studies pertained to the general and distressed populations. They cite one study that focused exclusively on college students. In light of the Healthy Minds Network data, there is a need to understand better any link between physical activity and positive aspects of mental health in college students.

A search in PubMed, EBSCO, SpringerLink, ProQuest, and Google Scholar for literature since the Zhang and Chen study yielded four articles that either focused on college students or the impact of physical activity on the positive attributes of mental health, but not both. Researchers in the U.S. analyzed whether certain types of leisure activities were associated with happiness and life satisfaction among 266 Indiana college students (Kim et al. 2023). In Spain, a cross-sectional study analyzed the relationship between physical activity and overall mental health in 847 undergraduate students (Rodriguez-Romo et al. 2023). Researchers in Poland examined whether initiating regular physical activity impacts happiness, life satisfaction, and self-esteem four weeks later through a longitudinal study in the general population (Iwon et al. 2021). German researchers undertook a longitudinal interventional study on university students to analyze the impact of short-term aerobic exercise on depression, anxiety, and perceived stress (Herbert et al. 2020). Due to a lack of studies that specifically focus on the effect of physical activity on college students' positive mental health attributes, the primary aim of this study was to examine the association between self-reported physical activity and happiness in college students.

2. Methods

2.1. Sample and Procedure

A descriptive cross-sectional survey was conducted using a Google Form questionnaire as a data collection instrument. The population was Washington University in St. Louis (WUSTL) undergraduate students. Data was collected from November 15, 2023, to November 28, 2023. A total of thirty-five surveys were collected. After reviewing the responses received, one survey was removed due to contradictory answers, and the final sample comprised thirty-four undergraduate students.

2.2. Measures

2.2.1. *Physical Activity*

The survey characterized physical activity as any movement that requires energy expenditure. Physical activity measures were collected with three questions, one to capture self-reported activity in each setting: Leisure Time, Commuting, and Occupational. Leisure Time activity was described as gym activities, swimming, sports, cross fit, or similar aerobic exercise. Commuting activity was described as physical movement to get to and from places like walking or biking to campus. Occupational activity was described as physical work as part of a participant's job. For each setting, respondents were provided with the choices of "Very Active: Daily," "Active: 5-6 times per week," "Moderate: 3-4 times per week," "Light: 1-2 times per week," and "Sedentary: Little to no activity." To quantify a participant's physical activity, points were allocated to those choices as 7, 6, 4, 2, and 1, respectively, corresponding to the weekly frequency of physical activity. Total physical activity, a measure designed to quantify the aggregate physical activity, was calculated as the sum of the activity points under each of the three settings. For each setting, the points were categorized as High, Moderate, or Low, depending on activity level. For the Total, scores representing the top third of aggregate activity points were categorized as High, the middle third points as Moderate, and the bottom third as Low. Respondents who indicated Sedentary for Leisure Time physical activity were queried as to what factors limit their physical activity

and given the options of “I don’t have enough free time,” “I would rather do something else with my free time,” or “Other” with the ability to provide free-form text. Respondents who indicated “Light” or above for Leisure Time physical activity were queried as to what factors motivated them to undertake aerobic exercise and given the options of “Getting/Staying fit,” “Looking and feeling better,” “Socializing,” “Improving Mood,” “De-stressing,” and “Other” with the ability to provide free-form text.

2.2.2. Happiness

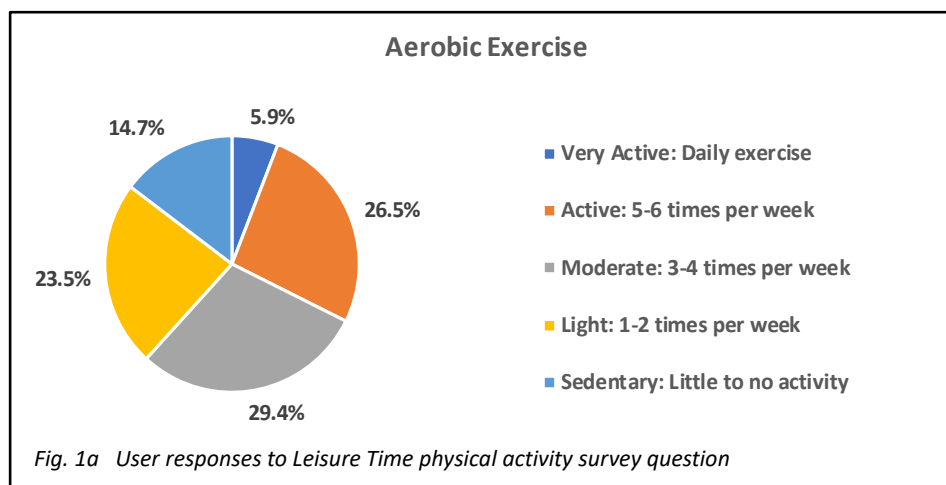
Happiness was measured with one question: “What is your current level of happiness (as an overall evaluation of life)?” The parenthetical expression in the question was included to minimize subjective interpretation of happiness, such as an assessment of the respondent’s happiness at the time of responding to the survey. The respondents were asked to pick one number between 1 and 10, with 1 representing “Very unhappy,” and 10 representing “Very happy.”

3. Results

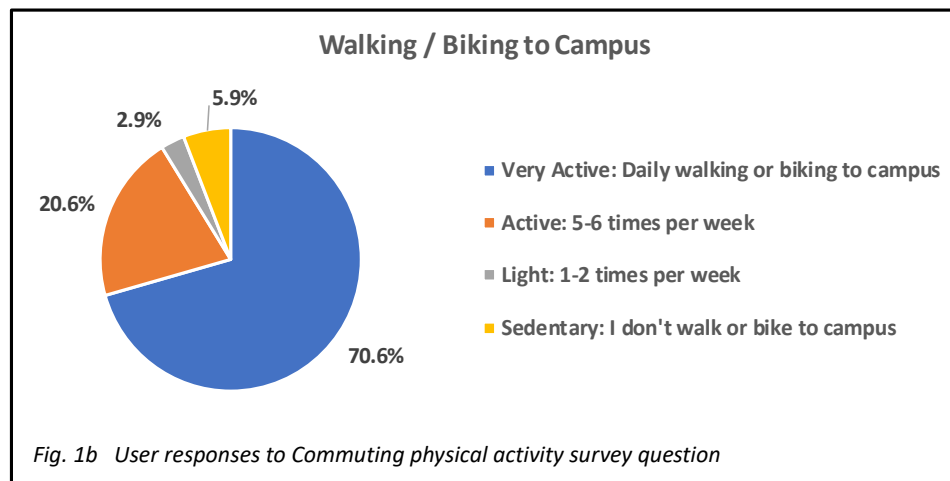
3.1. Data summary

3.1.1. Physical Activity data summary

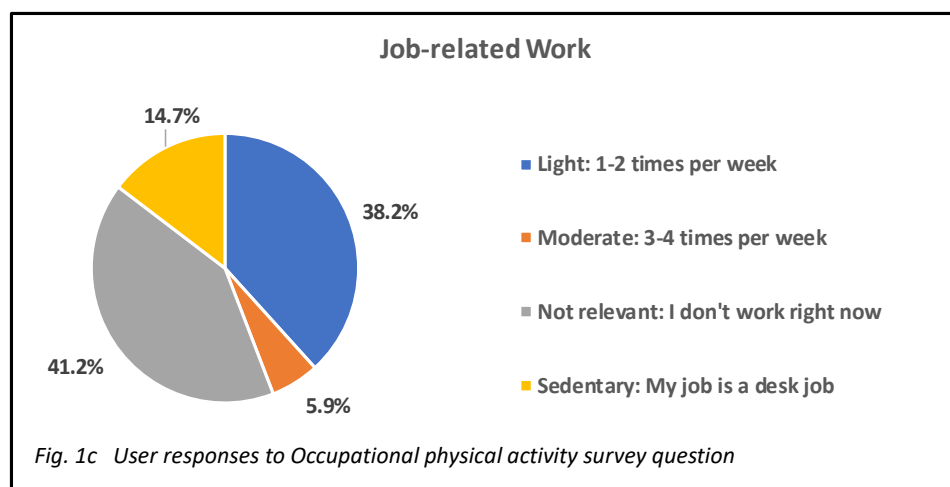
A summary of the responses to the Leisure Time physical activity question is shown in Fig.1a. 38% of the surveyed students indicated they are sedentary or light exercisers in their free time.



A summary of the responses to the Commuting physical activity question is shown in Fig.1b. 91% of the surveyed students indicated they walk or bike to campus nearly every day.



A summary of the responses to the Occupational physical activity question is shown in Fig.1c. Over half the surveyed students indicated they have jobs, which are sedentary or involve light physical activity.



The self-reported activity levels for each setting were converted to activity points using the conversion methodology described in 2.2.1, and the activity points under each setting was aggregated to calculate a Total physical activity metric for each participant. Also as described in 2.2.1, each participant's physical activity was then categorized as High, Moderate, or Low for each setting and for Total physical

activity. Table 1 shows a summary of the physical activity data under different settings and the categorization of participants following a conversion of self-reported descriptive activity to points.

Table 1 Physical activity data summary and categorization

Physical Activity Data and Categorization			
Setting	Participants	Points Range	Categorization
Total			
High	8	≥ 14	Top third of Total activity points
Moderate	12	11 - 13	Middle third of Total activity points
Low	14	≤ 10	Bottom third of Total activity points
Leisure Time			
High	11	6 - 7	Five or more days a week of aerobic exercise
Moderate	10	3 - 5	Three to four days a week of aerobic exercise
Low	13	1 - 2	Two days a week or less of aerobic exercise
Commuting			
High	24	7	Daily commute to campus
Moderate	7	3 - 6	Commute most but not all days to campus
Low	3	1 - 2	Commute 1 or 2 days a week to campus
Occupational			
High	2	≥ 3	Three or more days per week of job-related physical work
Moderate	18	1 - 2	One to two days per week of job-related physical work
Low	14	0	Sedentary job or currently not working

3.1.2. Happiness data summary

A summary of self-reported happiness scores is shown in Fig. 2. The mean happiness score for all participants was 7.5. The median and the mode were both 8 and the distribution shows a left skew.



Fig.2 Distribution of respondents' happiness score and summary happiness statistics

3.2. Correlation between happiness and physical activity and statistical significance

With students categorized as having High, Moderate, or Low physical activity, average happiness for participants in each category was computed in each of the settings and in total. A plot of the average happiness scores by activity level and setting is shown in Fig. 3.

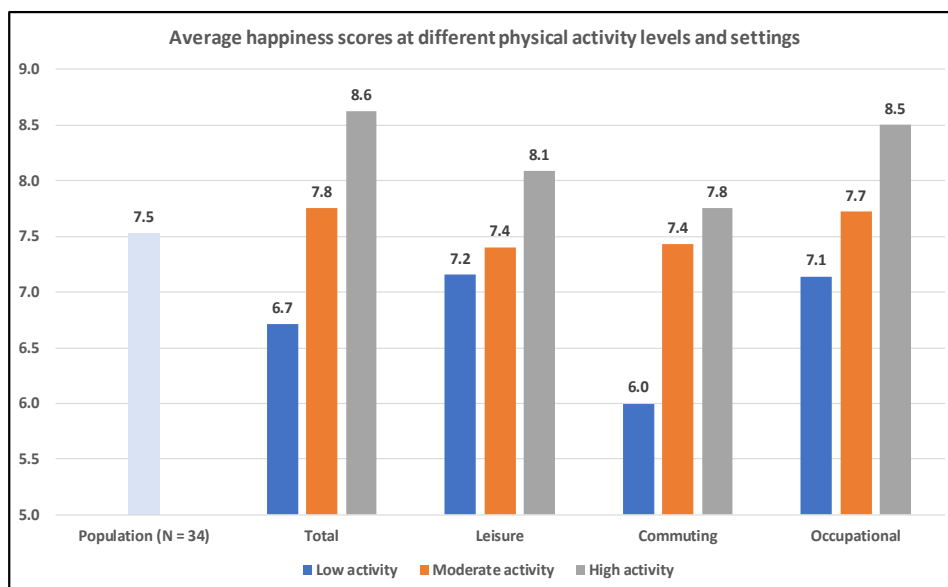


Fig.3 Average happiness scores at different activity levels and settings

To test whether the difference in average happiness between the High, Moderate and Low levels of Total physical activity observed in Fig.3 was statistically significant, an ANOVA single factor test was conducted at a significance level of 0.05. The results are shown in Table 2.

Anova: Single Factor

SUMMARY

Groups	Count	Sum	Average	Variance
High	8	69	8.625	0.553571
Moderate	12	93	7.75	0.931818
Low	14	94	6.714286	1.758242

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	19.48845	2	9.744223	8.168021	0.001415	3.304817
Within Groups	36.98214	31	1.192972			
Total	56.47059	33				

Table 2: ANOVA test for significance of difference in mean happiness between different levels of Total physical activity

The p-value of 0.001415, which is lower than the threshold of 0.05, indicates the difference in averages is statistically significant, and cannot be explained by chance. Similar tests for difference in average happiness between the High, Moderate and Low levels of physical activity in each of the settings did not indicate that the observed differences were statistically significant.

4. Discussion

This research analyzed the relationship between physical activity and happiness in undergraduate students, assessing whether the association can change depending on activity level (High, Moderate, or Low) and the setting in which it was performed (Leisure Time, Commuting, or Occupational). The study's results confirmed a statistically significant relationship between Total physical activity and happiness in undergraduate students. Specifically, the average happiness of students with High physical activity is higher than that of students with Moderate or Low levels of physical activity, and the average happiness of students with Moderate physical activity is higher than that of students with Low physical activity. Furthermore, the average happiness of students with High physical activity was higher than the average happiness of the overall population, and the average happiness of students with Low physical activity trailed that of the overall population.

The Leisure Time activity data suggests that the association between physical activity and happiness is not because those who exercise have more leisure time and hence are happier, as inferred from the fact that all participants except one who reported “Sedentary” for their Leisure Time physical activity indicated “I would rather do something else with my free time” in response to the query as to what factors limited their physical activity. Prior research suggests other physical and psychological mechanisms at work. Iwon et al. indicate that exercise releases endorphins that trigger positive emotions, important “components of happiness.” They also suggest that people who exercise regularly “may have a positive attitude toward their goals” and thus may have greater life satisfaction. Higher physical activity is also

linked with a more positive body image, which could result in greater self-esteem and happiness (Zamani Sani et al., 2016).

The study has limitations. Although the correlations between physical activity and happiness were also observed in the individual settings, statistical significance could not be established as with Total physical activity. In part, this could be the result of a skew in the sample towards first-year students. The sample comprised twenty-six first-year students, six sophomores, one junior, and one senior. Since first-year students live near campus and walk or bike to work, the data for physical activity under the Commuting setting was skewed heavily by the concentration of first-year students in the sample. Most respondents indicated they walk or bike to campus at least five days a week. As a result, there was not enough variability to detect a difference in happiness. The high proportion of first-year students also impacted the data for physical activity in the Occupational setting, as many students do not take on jobs in their first year. Also, those students who did work had jobs that were sedentary or only involved light physical work. As a result, there was not enough variability to detect a difference in happiness. The study used a simple one-question format to gather physical activity data in each setting and quantified the activity based on frequency regardless of duration or intensity. For example, a participant who goes to the gym twice a week for two hours each gets only half the points as another participant who goes four times a week for an hour each, even though they both exercise for the same duration. A similar point applies to the intensity of exercise. Finally, the study's cross-sectional nature prevents establishing causal relationships between physical activity and happiness.

There are several opportunities for enhancing the study. Physical activity and happiness can be measured using internationally standardized questionnaires that reduce variability due to different interpretations of questions. The student sample set can be crafted to include a more balanced representation between first-years, sophomores, juniors, and seniors. More randomized controlled trials are warranted for future research to draw causal conclusions.

At WUSTL, there is an opportunity to increase to increase students' happiness. 38% of students indicated they are sedentary or have low physical activity during their free time. Many colleges have a physical education requirement to graduate. While mandating physical activity may not be the best policy, other options exist to encourage students to exercise. For example, the Habib Health and Wellness Center offers WashU Recreation Center members BearFIT classes in cycling, boxing, yoga, and other physical activities. WUSTL management could consider offering such courses for credit. Sedentary students interested in such activity may be motivated to enroll if they can earn credits while engaging in a fun physical activity.

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